

Tender Management System (TMS)

Full-Stack Architecture & File Segregation Guide |

Jakarta EE 10

Servlet 6.0

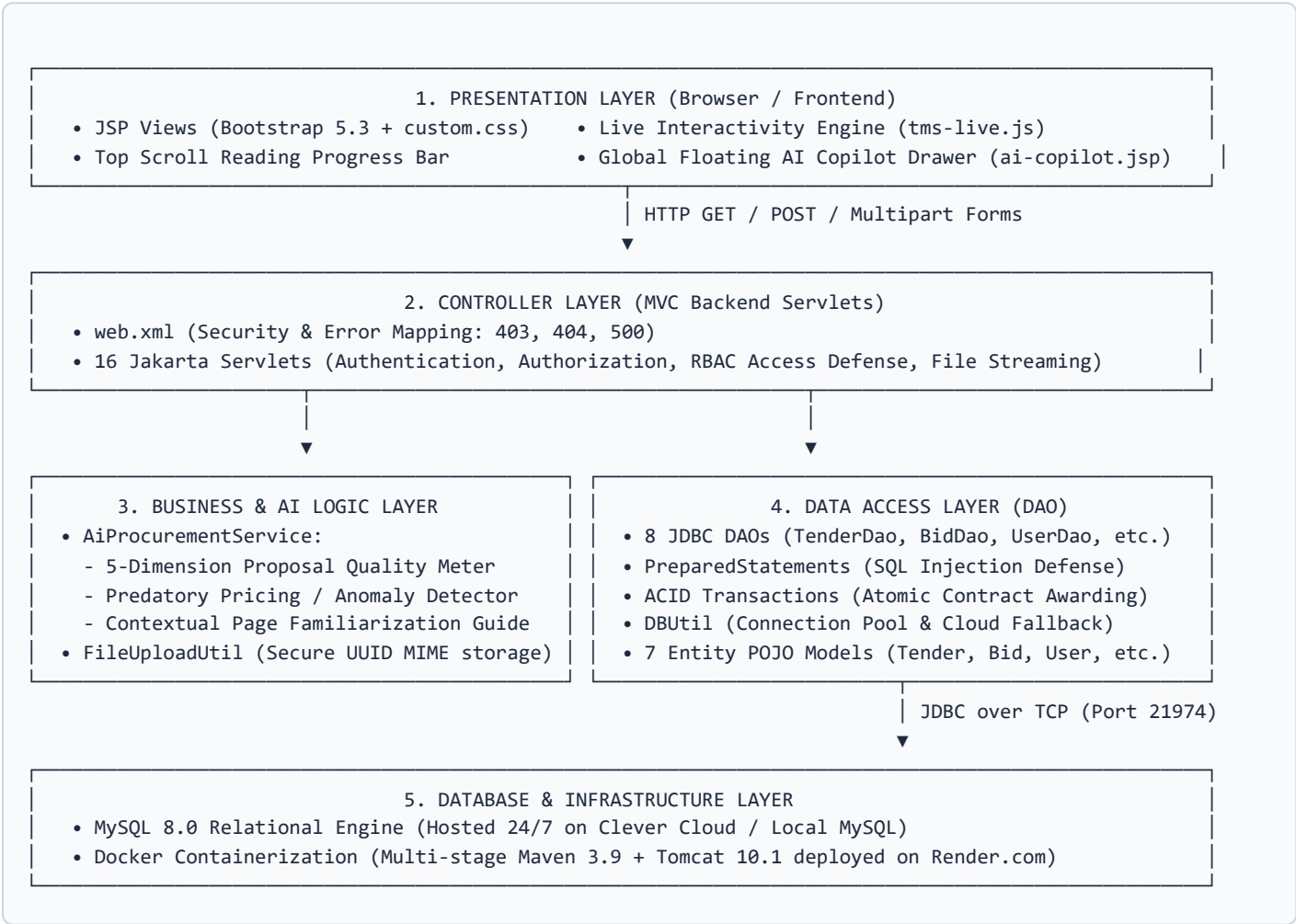
MySQL 8.0

Apache Tomcat 10.1

Docker / Render

System Overview: An enterprise 3-tier Java web application featuring role-based access control (Admin & Vendor), offline heuristic AI proposal evaluation, live ticking deadline clocks, and automated L1 contract awarding with official legal certificate generation.

1. High-Level Architecture Flowchart



2. Master File Segregation Chart

Layer 1: Presentation / Frontend Layer (src/main/webapp/)

File / Path	Architectural Role	What It Does (Simple Explanation)
css/custom.css	Global Design System	Styles the glassmorphic navbar, typography, micro-elevations, animations, and PDF print rules.

File / Path	Architectural Role	What It Does (Simple Explanation)
<code>js/tms-live.js</code>	Client Interactivity Engine	Drives live countdown clocks, instant table filtering, count-up metric tickers, and toast alerts.
<code>login.jsp</code>	Authentication View	Renders the sign-in form with 1-click Admin Demo and Vendor Demo filler buttons.
<code>signup.jsp</code>	Onboarding View	2-column contractor registration portal with real-time password matching feedback.
<code>index.jsp</code>	Root Session Router	Inspects active session and safely routes users to <code>/dashboard</code> , <code>/tenders</code> , or <code>/login.jsp</code> .
<code>navbar.jsp</code>	Shared Navigation Header	Header with role-aware menus, brand shield emblem, and live in-app notification bell.
<code>ai-copilot.jsp</code>	Global Assistant Drawer	Floating offcanvas drawer providing contextual page walkthroughs and conversational Q&A.
<code>dashboard.jsp</code>	Executive Admin View	Displays 6 animated KPI cards, recent solicitations table with instant filter, and copy chips.
<code>tenders.jsp</code>	Catalog Directory View	Searchable tender directory with Quick Sector pills (Solar, Infra, IT) and countdown timers.
<code>tender-detail.jsp</code>	Specification View	Displays RFP requirements, scope downloads, copy chips, and AI executive summary card.
<code>tender-form.jsp</code>	Procurement Authoring	Form for admins to create and edit tenders with PDF/DOCX specification file uploads.
<code>bid-form.jsp</code>	Commercial Quotation	Vendor bid submission with live word counter, AI Auto-Draft , and AI Quality Meter .
<code>bid-comparison.jsp</code>	Evaluation Board View	Ranks quotes from lowest (L1), featuring the AI Bid Anomaly & Risk Detector .
<code>my-bids.jsp</code>	Vendor Portfolio View	Displays submitted bids, proposal preview modals, and direct links to award certificates.
<code>vendor-approvals.jsp</code>	Compliance Queue	Admin verification queue to verify contractor registrations and approve/reject with 1 click.
<code>award-certificate.jsp</code>	Legal Contract Document	Official formal certificate with dual signatures, security watermark, and 1-click PDF print.
<code>notifications.jsp</code>	Alert History Stream	Displays chronological notifications with unread badges and "Mark all as read" button.
<code>audit-logs.jsp</code>	Regulatory Audit View	Displays chronological, immutable compliance history of all system activities.
<code>access-denied.jsp</code>	Security Exception View	Branded HTTP 403 Forbidden page shown when a vendor tries accessing admin areas.
<code>error-404.jsp</code>	Routing Exception View	Branded HTTP 404 Not Found page for non-existent URLs.

File / Path	Architectural Role	What It Does (Simple Explanation)
<code>error-500.jsp</code>	Server Exception View	Branded HTTP 500 Server Error page for unexpected backend exceptions.

Layer 2: Controller & Routing Layer (`src/main/java/com/tms/servlet/`)

Servlet Class	URL Pattern	What It Does (Simple Explanation)
<code>LoginServlet.java</code>	<code>/login</code>	Verifies credentials, generates session tokens, and routes users to appropriate landing page.
<code>LogoutServlet.java</code>	<code>/logout</code>	Destroys active session attributes and safely redirects user back to <code>login.jsp</code> .
<code>RegisterServlet.java</code>	<code>/register</code> , <code>/signup</code>	Validates contractor registration, password matching, and executes transactional insertion.
<code>DashboardServlet.java</code>	<code>/dashboard</code>	Fetches high-level procurement metrics and blocks non-admin users with HTTP 403.
<code>TenderListServlet.java</code>	<code>/tenders</code>	Processes catalog search queries, category filters, and tender status checks.
<code>TenderDetailServlet.java</code>	<code>/tender</code>	Loads tender specifications and checks if the active vendor has already submitted a bid.
<code>TenderFormServlet.java</code>	<code>/tender-form</code>	Processes admin tender publishing, updates, and handles specification file uploads.
<code>BidServlet.java</code>	<code>/bid</code>	Enforces vendor approval, blocks duplicate bids, validates amount, and saves proposal PDF.
<code>BidComparisonServlet.java</code>	<code>/bids</code>	Handles L1 comparison and executes atomic contract awards across multiple tables.
<code>MyBidsServlet.java</code>	<code>/my-bids</code>	Loads vendor-specific bids and portfolio summary KPIs.
<code>VendorApprovalServlet.java</code>	<code>/vendor-approvals</code>	Lets administrators approve or reject pending contractor applications.
<code>FileDownloadServlet.java</code>	<code>/download</code>	Streams attachments with directory traversal guards and competitor proposal blocking.
<code>AwardCertificateServlet.java</code>	<code>/award-certificate</code>	Validates contract award and serves printable certificate only to Admin and winner.
<code>NotificationServlet.java</code>	<code>/notifications</code>	Serves notification list and delivers JSON streams (<code>?format=json</code>) for navbar polling.
<code>AuditLogServlet.java</code>	<code>/audit-logs</code>	Serves paginated compliance history to administrators.
<code>AiAssistantServlet.java</code>	<code>/api/ai</code>	Handles asynchronous AJAX calls for page tours, chat questions, and proposal scoring.

Layer 3: Business & Domain Intelligence Layer (`src/main/java/com/tms/service/`)

Service Class	Technical Role	What It Does (Simple Explanation)
<code>AiProcurementService.java</code>	Local Embedded AI Engine	Executes rule-based NLP algorithms for proposal scoring, risk checks, and page familiarization without external APIs.
<code>FileUploadUtil.java</code>	Secure File Handler	Enforces file extensions (PDF/DOCX), generates unique UUID filenames, and prevents path traversal attacks.

Layer 4: Data Access & Entity Layer (`src/main/java/com/tms/dao/` & `model/`)

Component	Technical Role	What It Does (Simple Explanation)
<code>TenderDao.java</code>	SQL Persistence Handler	Executes SQL for tenders (creating, editing, status changes, multi-filter searching).
<code>BidDao.java</code>	SQL Persistence Handler	Executes SQL for bids, prevents duplicate bids (<code>hasVendorBid</code>), and fetches quotes.
<code>ContractAwardDao.java</code>	Transactional Coordinator	Runs atomic SQL transactions (awards winning bid, rejects losing bids, closes tender).
<code>VendorDao.java</code>	SQL Persistence Handler	Handles vendor lookups, approval updates, and CIN uniqueness checks.
<code>UserDao.java</code>	SQL Persistence Handler	Manages user credentials, authentication lookups, and account queries.
<code>DashboardDao.java</code>	Analytics Aggregator	Executes optimized aggregate SQL queries (<code>COUNT</code> , <code>SUM</code>) to feed dashboard KPI cards.
<code>NotificationDao.java</code>	Alert Persistence Handler	Creates system alerts, tracks unread counts, and updates read states.
<code>AuditLogDao.java</code>	Immutable Event Logger	Writes tamper-proof audit trails for every critical administrative action.
<code>DBUtil.java</code>	Connection Pool Manager	Provides JDBC connections, reading cloud environment variables (<code>DB_URL</code> , <code>DB_USER</code> , <code>DB_PASSWORD</code>).
<code>Model Classes (*.java)</code>	Data Transfer Objects	<code>Tender</code> , <code>Bid</code> , <code>User</code> , <code>Vendor</code> , etc., holding data in typed variables with getters/setters.

Layer 5: Database Layer (`database/`)

SQL Script	Technical Role	What It Does (Simple Explanation)
<code>schema.sql</code>	Database DDL Blueprint	Creates the 7 relational tables with primary keys, foreign keys, indexes, and constraints.

SQL Script	Technical Role	What It Does (Simple Explanation)
<code>seed.sql</code>	Initial Data DML Script	Populates initial demo accounts (<code>admin@tms.com</code> , <code>vendor1@tms.com</code>), vendors, and sample tenders.

Layer 6: Build & Cloud Deployment Layer (Root Folder)

Configuration File	Technical Role	What It Does (Simple Explanation)
<code>pom.xml</code>	Maven Build Descriptor	Defines project dependencies (<code>jakarta.servlet-api</code> , <code>jstl</code> , <code>mysql-connector-j</code>) and packages the <code>.war</code> .
<code>Dockerfile</code>	Multi-Stage Container Recipe	Builds the WAR with Maven in Stage 1, and deploys it to Tomcat 10.1 in Stage 2 as <code>ROOT.war</code> .
<code>.dockerignore</code>	Build Optimization	Excludes temporary logs, local classes, and test files from being uploaded to Docker.
<code>render.yaml</code>	Infrastructure Blueprint	Tells Render.com to spin up a Docker container on port 8080 and connect the database.
<code>start-tms.bat</code>	Local Server Launcher	1-click Windows shortcut to launch local Apache Tomcat 10.1 without terminal commands.
<code>push-to-github.bat</code>	Version Control Launcher	1-click script to push any local code changes straight to your GitHub repository.

 3. Summary of How the Stack is Segregated

- 1. **Frontend** (`src/main/webapp`): Pure UI rendering (JSP), styles (`custom.css`), and browser interactivity (`tms-live.js`). Zero business logic.
- 2. **Backend** (`src/main/java`): The brain. Servlets handle security and routing; DAOs talk safely to the database via `PreparedStatement` ; AI services run scoring algorithms. Zero HTML generation.
- 3. **Database** (`database/`): Relational storage engine (MySQL). Tables, foreign keys, and permanent records.
- 4. **Deployment (Root)**: Docker and Maven packaging enabling the system to run on any computer or cloud host with zero manual setup.